

Radar Project Proposal

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1 Introduction

Detecting and tracking humans behind obstacles is a critical capability for applications such as search-and-rescue missions, security surveillance, and smart home automation. Traditional vision-based systems fail when visibility is obstructed by walls or furniture, whereas mmWave Frequency Modulated Carrier Wave (FMCW) radar can penetrate obstacles and provide tracking. In this project, we aim to develop an efficient through-wall human detection and counting system using mmWave radar and deep learning techniques.

As an extra feature, we aim to use breathing analysis to estimate the average stress level of people in the room. This may allow us to classify the environment as calm or active which could be valuable in contexts like emergency response, classroom behavior monitoring, or crowd control.

2 Project Plan

1. Radar Data Processing: Capture and preprocess mmWave radar signals to extract meaningful patterns.
2. Heatmap Generation: Convert radar reflections into spatial heatmaps representing human presence. Develop algorithms to detect and count individuals within a room.
3. Machine Learning for Shape Reconstruction: Develop a deep learning model to reconstruct human shapes from heatmaps.
4. Efficient Real-Time Tracking (Extra Feature): Optimize the pipeline for real-time performance with minimal latency.
5. Skeleton Extraction (Extra Feature): Use confidence maps to infer a person's skeletal structure.
6. Emotion Recognition (Extra Feature): Investigate the feasibility of detecting emotions based on subtle breathing patterns.

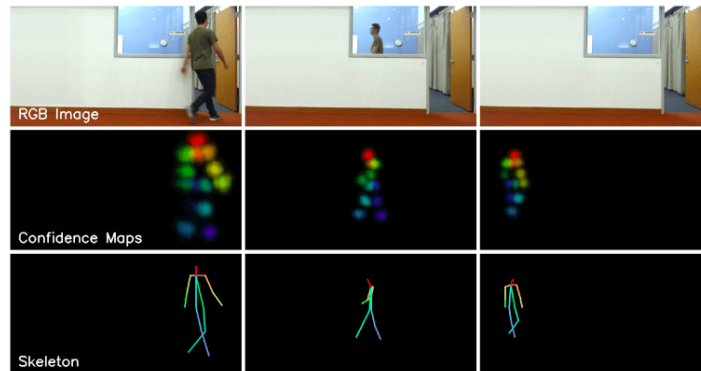


Figure 1: Image source : [nbcnews.com](https://www.nbcnews.com)

3 Technical Approach

3.1 mmWave Radar Setup

1. Hardware: Texas Instruments (TI) AWR1843BOOST mmWave FMCW radar, 24GHz - 77GHz, combined with the DCA1000EVM data capture board.

3.2 Approach

1. Signal Processing:
 - Capture raw mmWave radar data and perform pre-processing.
 - Extract Range-Doppler and Range-Azimuth heatmaps using Fast Fourier Transform (FFT) and beamforming algorithm for efficiency.
 - Implement background subtraction to filter out static objects.
 - Use adaptive thresholding techniques to detect dynamic objects while reducing noise.
 - Grouped People Counting: [Download Paper \(TU Delft\)](#)
2. Heatmap and Confidence Map Generation
 - Convert processed radar signals into 2D heatmaps, highlighting intensity variations that correspond to human presence.
 - We will be using at ≥ 2 radars, to combine X-Y coordinates data to triangulate positions and separate overlapping individuals.
 - Apply Constant False Alarm Rate (CFAR) detection to enhance target detection while minimizing false positives.
 - Use clustering techniques (K-Means) to distinguish multiple moving objects and track them independently.
 - Generate confidence maps by assigning probability values to detected regions based on their signal strength and consistency over time.
 - Grouped People Counting: [Download Paper \(TU Delft\)](#)
3. Deep Learning-Based Shape Reconstruction
 - Train a Convolutional Neural Network (CNN) model to learn the correlation between radar heatmaps and human shapes.
 - Leverage pre-existing datasets (human motion datasets captured with LIDAR or cameras) to train the network.
 - Augment training data by simulating radar reflections of different body postures and environments.
 - People Counting using CNN: [GitHub: cristinaa211/People-counting-CNN-UWB](#)
4. Real-Time Optimization (Extra Feature)
 - Deploy the trained deep learning models using TensorRT or ONNX Runtime for high-speed inference.
 - Optimize data pipelines to minimize processing latency, ensuring real-time updates at ≥ 30 FPS.
 - mmWave-Based People Tracking: [GitHub: AsteriosPar/mmWave_MSc](#)
5. Skeleton Extraction and Tracking (Extra Feature)
 - Convert reconstructed human shapes into pose confidence maps, where key body parts (head, hands, feet) are assigned probability scores.
 - Implement a Pose Estimation Algorithm (OpenPose-inspired model) to infer skeletal joints from the confidence maps.

- Use Kalman Filters or Particle Filters to smooth tracking and predict movements.
 - OpenMMLab’s Pose Estimation Toolbox: [GitHub: open-mmlab/mmpose](#)
6. Emotion Recognition via Breathing Analysis (Extra Feature)
- Use beamformed micro-motion data to isolate periodic chest movements.
 - Use FFT on signal windows to detect breathing rate variations.
 - Train a classifier (e.g., CNN or SVM) to distinguish between calm and stressed states based on respiration frequency and depth.
 - Compute the average stress index of all detected individuals in real-time.
 - Use the aggregate stress index to categorize the environment as:
 - Passive Room – Mostly calm, low movement, relaxed breathing patterns.
 - Active Room – High stress, intense movement, fast breathing.
 - Household Radar, Seeing Through Walls and Detecting Emotions: [IEEE Spectrum Article](#)

4 Roadmap

Week	Milestone
5	Project Proposal Submission
6	Radar setup & Radar Signal Processing Implementation
7–8	Heatmap and Confidence Map Generation
9	Heatmap and Confidence Map Generation & Project Progress Due
10	Break
11–12	Machine Learning Model Development
13	Real-Time Optimization
14	Extra features
15	Final Testing & Report Submission

5 Risks

This project faces several key risks:

1. **Radar Data and Signal Processing** : Poor data quality (e.g., noise, miscalibration) and the complexity of signal processing algorithms can lead to unreliable detection if not properly optimized.
2. **Occlusion** : When one person blocks another, or when several people are very close to each other, the radar may detect a merged signal. Which would leading to inaccurate counting or shape estimation, especially in dense environments.
3. **Real-Time Constraints** : Hardware limitations may hinder fast processing, making it difficult to achieve smooth, real-time tracking without effective optimization techniques.
4. **AI Model Accuracy** : An insufficient or low-quality training dataset can reduce the model’s ability to accurately detect and track human presence.

6 Related Work and Resources

Here are some useful references we plan to use for our project:

- **Grouped People Counting Using mmWave FMCW MIMO Radar**
[Download Paper \(TU Delft\)](#)
- **People Counting using CNN and UWB Radar (GitHub Repo)**
[GitHub: cristinaa211/People_counting_CNN_UWB](#)
- **mmWave-Based People Tracking (MSc Thesis)**
[GitHub: AsteriosPar/mmWave_MSc](#)
- **MMPose: OpenMMLab's Pose Estimation Toolbox**
[GitHub: open-mmlab/mmpose](#)
- **Household Radar: Seeing Through Walls and Detecting Emotions**
[IEEE Spectrum Article](#)